The A.C. Generator

Exam code: 0625 & 0972

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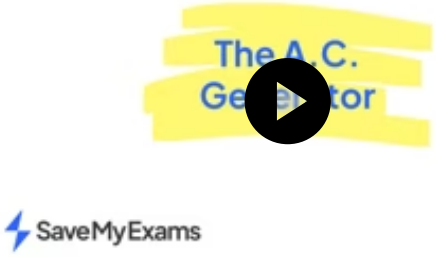

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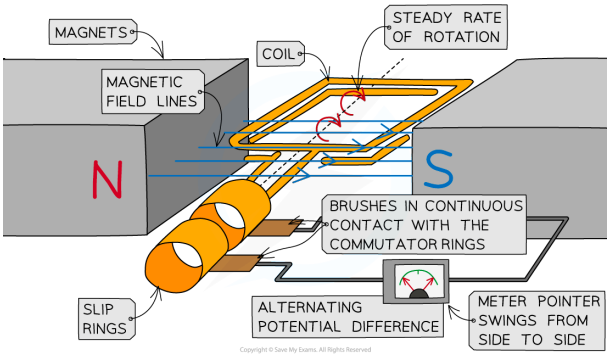
Simple a.c. generators

Extended tier only

- An a.c. generator is a device which converts energy from **motion** into an **electrical** output
- An alternating e.m.f. is generated which causes an **alternating current** to flow
- A simple a.c. generator consists of
  - a rotating coil of wire between the poles of a permanent magnet
  - slip rings and brushes connected to an external circuit

Structure of a simple a.c. generator



Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The functions of each component are shown in the table:

Table of components of a simple a.c. generator

Component	Function
permanent magnet	to provide a uniform magnetic field
rotating coil	to cut the magnetic field as it rotates and allow an induced current to flow
slip rings	to allow the alternating current to flow between the coil and the external circuit
carbon brushes	to provide a good electrical connection between the coil and the external circuit

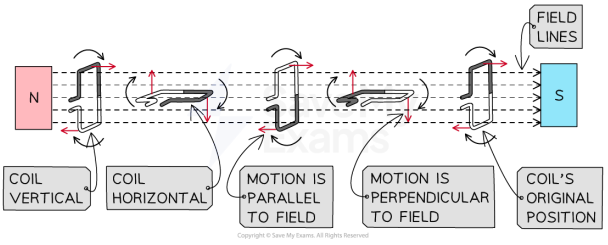
Operation of an a.c. generator

- A rectangular coil rotates in a **uniform magnetic field**
- The coil is connected to an external circuit via **slip rings and brushes**
  - The induced emf in the coil can be measured by adding a galvanometer (centre-zero meter) to the external circuit

- An e.m.f. is **induced** in the coil as it **cuts** the magnetic field
  - The pointer defects first one way, then the opposite way, and then back again
  - This indicates the size and direction of the emf is constantly changing

- As a result of the alternating e.m.f., an alternating **current** is also produced as the coil rotates
  - This continues as long as the coil keeps turning in the **same** direction

Motion of an a.c. generator

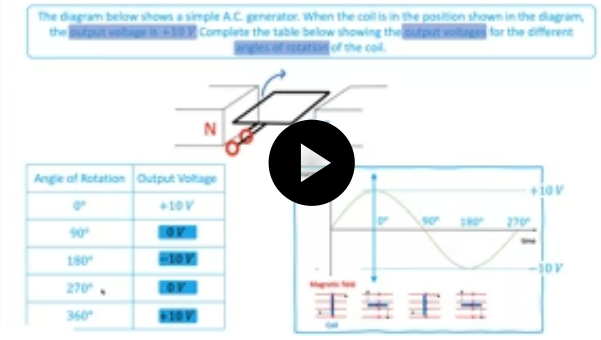


The size and direction of the induced e.m.f. (and current) depend on the orientation of the coil with the field

- A maximum e.m.f. is induced when
  - the **position** of the coil is **parallel** to the field

the coil is moving perpendicular to the field

- No e.m.f. is induced when
  - the **position** of the coil is **perpendicular** to the field
  - the **motion** of the coil is **parallel** to the field
- This is because no lines are cut when the coil is moving parallel to the field



Did this video help you? Yes No

Graphs for a.c. generators

Extended tier only

- The output of an a.c. generator can be seen on a graph of e.m.f. against time, or angle of rotation
- The shape of the graph is a sine or cosine curve, depending on the starting position of the coil
  - When it starts from a **horizontal** position (e.m.f. is at a maximum), the graph is a **cosine** curve
  - When it starts from a **vertical** position (e.m.f. is zero), the graph is a **sine** curve

Graph of induced e.m.f. with angle for an a.c. generator

Alternating e.m.f. with corresponding positions of the coil relative to the field

- When the coil is vertical at 0°
  - it is moving **parallel** to the direction of the magnetic field
  - the size of the induced e.m.f. is **zero**
- When the coil has rotated by 90°
  - it is now **horizontal** and moving **perpendicular** to the direction of the magnetic field
  - the size of the induced e.m.f. is at a **maximum**
- When the coil has rotated by 180°
  - it is **vertical** again and moving **parallel** to the direction of the magnetic field
  - the size of the induced e.m.f. is **zero**
- When the coil has rotated by 270°
  - it is **horizontal** again and moving **perpendicular** to the direction of the magnetic field
  - the size of the induced e.m.f. is at a **maximum** and in the **opposite** direction to its position at 90°
- When the coil has completed a full 360° rotation
  - it is back at its starting point where it is moving **parallel** to the direction of the magnetic field
  - the size of the induced e.m.f. is **zero**

Factor affecting a.c. generators

- The magnitude of the induced e.m.f. can be increased by:
  - increasing the **frequency** of rotation of the coil
  - increasing the number of **turns** on the coil
  - increasing the **strength** of the magnet
  - inserting a **soft iron core** into the coil

Examiner Tips and Tricks

- a **coil rotating** in a magnetic field
- a **magnet rotating** within a coil

Both will induce an e.m.f. in the coil as they both ensure the coil will experience a **changing** magnetic field.

Take a look at these notes on trigonometric graphs if you need to brush up on your knowledge of sine and cosine graphs.

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